

ENME302 S2 2026



Course Handbook - Volume I Finite Element Analysis Computational and Applied Mechanical Analysis

Semester Two

Te Tari Pūhanga Pūrere | Mechanical Engineering

ENME302-26S2-HBK-VOL1

Te Kaupeka Pūhanga | Faculty of Engineering

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ENME302-26S2

Computational and Applied Mechanical Analysis

Semester 2

2026

Mechanical Engineering

Finite Element Analysis

Week 1-4

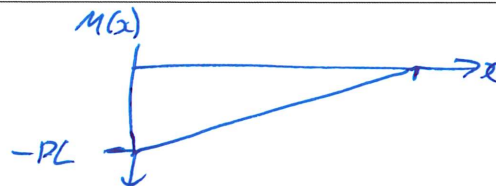
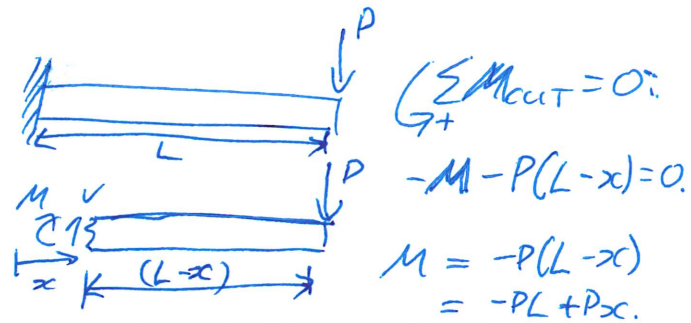
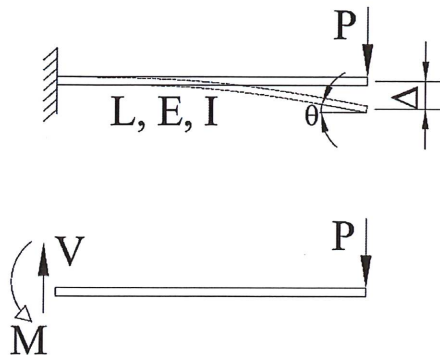
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E511, 5th Floor of the Civil/Mech Building

Finite element analysis

Scenario 1

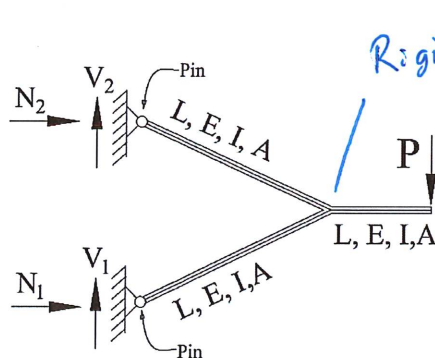


Question:

- What is deflection, Δ ?
- What are reaction loads V , M ?

From ENME202 we know $\Delta_{tip} = \frac{PL^3}{3EI}$, $\theta_{tip} = \frac{PL^2}{2EI}$, $V = -P$, $M = -PL$

Scenario 2



Rigid connection at the central node where elements meet.

Question:

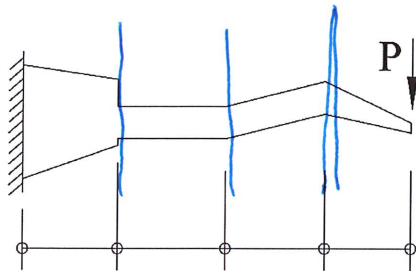
- What is the end deflection at load point, P ?
- The are reaction forces, V_1 , V_2 , N_1 , N_2 ?

...And it gets harder from there, especially when we consider "elements" that carry moment (beams) instead of purely axial load carrying elements (bars).

The Solution: use Finite Elements (FE) to break the problem into simpler pieces and simplify solution.

- ⇒ Break a large, complex structure into a number of smaller, simpler pieces, where each piece can be modelled easily, and can together be used to model a range of possible structures.
- ⇒ Develop a framework under which these simple, basic, elements can be combined together to represent the larger, more complicated, structures.

Scenario 3



Question:

- How to calculate the end deflection of this complex element at load point, P?

Solution → break the element into at least 4 unique finite elements with unique cross-sections.

Almost any structure can be broken down into simpler pieces.

⇒ Underlying principle of Finite Element Analysis.

The Finite Element Method consists of the following steps:

Week 1-2

A. Choosing the element type

- Axial trusses- carry only axial forces (tension/compression), but do not carry any moments or shear forces.
- Flexural beam- long and slender, carry moments and/or shear forces
- Shear beam- short and squat, carry moments and/or shear forces

Weeks 3-4.

B. Identifying boundary conditions

- Fixed
- Free
- Constrained – pinned, simply-supported (knife-edge), rollers, slotted connections.

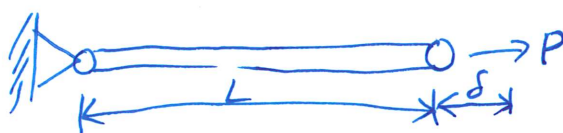
C. Applying forces

- Point forces
- Moments
- Continuous (spread) loads

D. Choosing degrees of freedom

- Locations where deflections are to be solved for directly

Axial Bar: Pinned at each end, loads are only applied at nodal points.



$$\delta = \frac{PL}{AE}$$

$$P = \left(\frac{AE}{L}\right) \delta$$

→ this term represents stiffness.

$F = Kx$ (Hooke's Law equation for a spring)

The "Finite Element Method" (FEM)

Again, any structure can be broken down (discretized) into simpler elements and be analysed using finite element method. The problem can be formulated and analysed as follows:

Problem formulation:

- 1) Approximate exact force-displacement behaviour of each piece:

The fundamental equation that relates applied loads to the resulting deflections.

$$f^e = K^e d^e$$

K^e - stiffness matrix

f^e - a vector of applied loads

d^e - a vector of allowed deflections

Hooke's Law equation but applied in a matrix sense.

- 2) Assemble individual stiffness equations ($f^e = K^e d^e$) into a larger model ($\hat{F} = \hat{K} \hat{D}$)

Transform and combine the individual elements into a larger more complex structure.

- 3) Apply constraints to remove DOF that are fixed or known (motions) to get a final model

$$(F = KD).$$

~ Apply boundary conditions.

- 4) Solve the deflections for the given applied loads: $D = K^{-1}F$

*matrix inverse
or linear algebra solution.
np.linalg.solve.*

Analysis of results:

- 5) Using this solved deflection vector, D , to calculate reaction forces (V,N,M)
6) From D as we will see you can also estimate/approximate all motion within an element (not just at the 'nodal' or 'end' points).

→ Overall, you can solve very difficult problems using the "Finite Element Method" (FEM).

We will cover:

1) Bar (or truss) elements:

- Structures made up of bar elements are assumed to have pinned connections between adjacent/connecting elements, and that elements are pinned to any support points.
- Loads are applied only at nodal points.
 - As a result, each Bar Element carries only a **constant axial load** (no moments or shear forces).

2) Beam elements:

- carry moment, shear forces, lateral displacements and rotations
- can have shear motion perpendicular to the length of element

For each case we will:

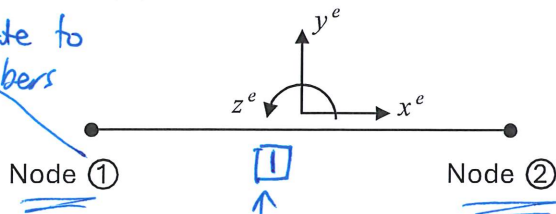
- 1) Demonstrate element formulation $f^e = K^e d^e$.
 - 2) Discuss rotation of the coordinate system
 - 3) Demonstrate assembly of many elements into a single model for solution
 - 4) Work through different examples and develop Codes to solve.
- ⇒ Then, if you are using a commercial finite element package, such as ANSYS, ComSol, Abaqus, etc, it's not just a "black box" where you turn a handle and numbers come out. Instead, you have an idea of what's going on "under the bonnet" and can therefore better understand when something goes wrong, and a better understanding of the limitations in the analysis method.

Nomenclature

This course will use the following nomenclature to describe analysed structures:

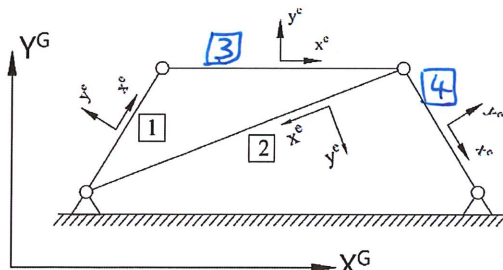
1. Element:

Circles relate to node numbers within an element.



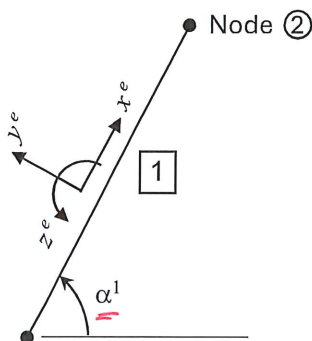
Squares around a number relate to element number.

2. Global Model



Global coordinate system.

Element 1



Node 1

- α^1 corresponds to angle going counter-clockwise (CCW) from $X^G \rightarrow x^e$ for element 1

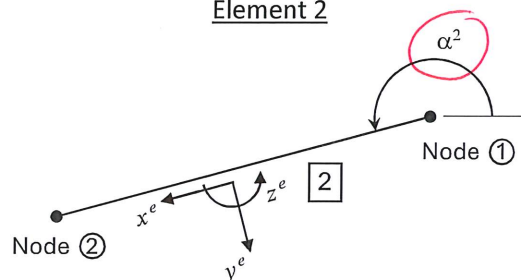
Element 1

- (x^e, y^e) – element coordinate system
- x^e is always defined along the length of element from node 1 towards node 2.
- (x^e, y^e) are always in the plane of the page, which defines z^e as upwards out of the page with a vector that is normal to the plane of the page.
- We show z^e as a counter-clockwise (CCW) rotation, defined by the right-hand rule.

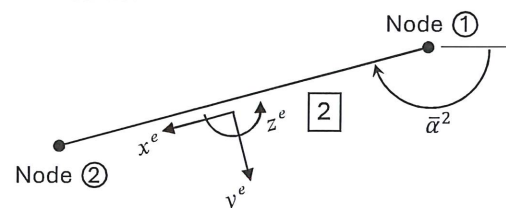
- You can pick any end as “node 1” as long as you align x^e going from node 1 towards node 2.
- (X^G, Y^G) – global coordinate system. Set for convenience. Usually, X^G is set along the ground (horizontal) and Y^G is vertical as this is usually the most convenient.

Element orientation angle, α .

Element 2



- α^2 corresponds to angle measured counter-clockwise from X^G to x^e
- In this case, you could use a transformation angle of $\alpha^2 = 195^\circ$ or $\bar{\alpha}^2 = -165^\circ$ and the result will be the same.



Terminology:

“Displacements” or **“Translations”** – the linear component of deflection, measured in metres (m) or millimetres (mm).

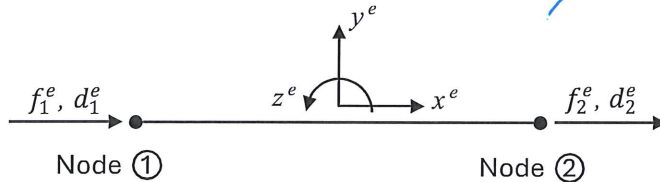
“Rotations” – The result of shear deformations and moments, measured in radians (rad) or milliradians (mrad).

“Deflections” – A more general term for the way in which an element can move or deform and can include both **Displacements/Translations** and **Rotations**.

Overview: What the finite element method tries to accomplish

(1) A simple structure

- Assume one element:



Simple axial element,
constant A , constant E
can carries a constant
axial load.

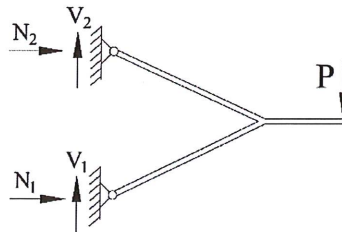
- Analytical solution is easy enough
- We can create an element stiffness matrix K^e relating f_1^e and f_2^e to d_1^e and d_2^e

$$f^e = K^e d^e$$

- We will define $u(x)$ as some approximation to the easy analytical solution knowing only the values at the two nodal points at each end of the element, $u(x=0) = d_1$ and $u(x=L) = d_2$

~ Boundary Value Problem.

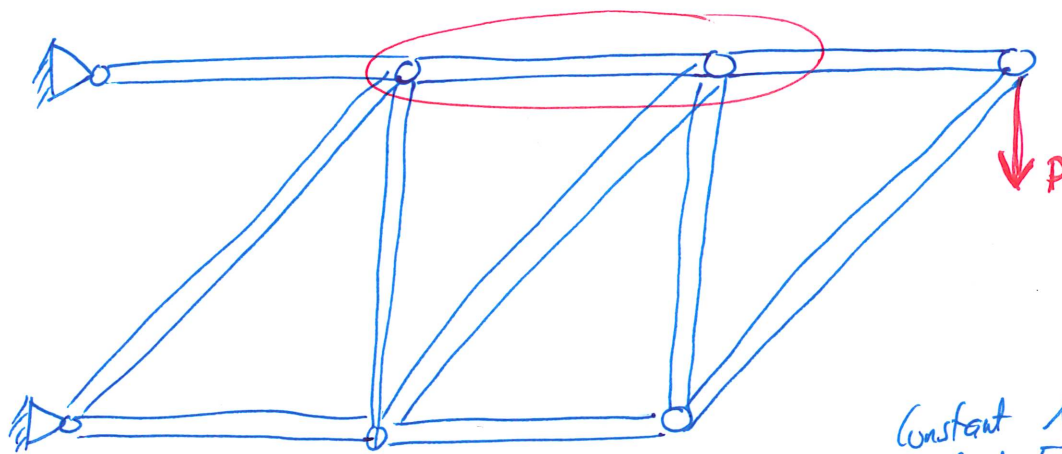
(2) A (reasonably) complex structure



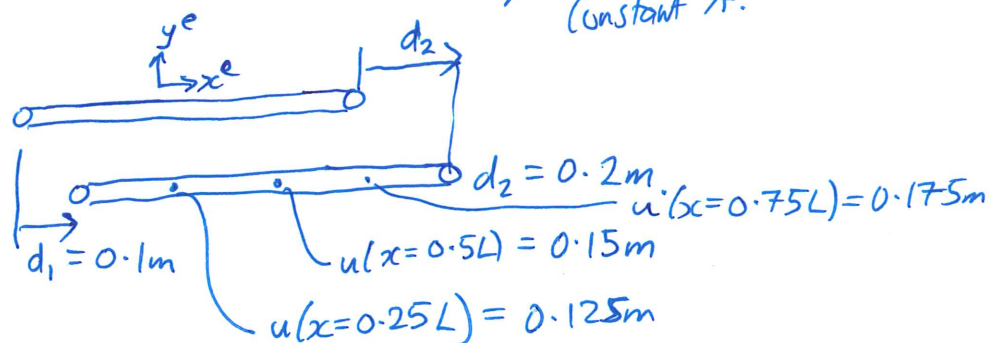
Given 3 elements that comprise the structure, analytical solution may no longer be possible. Hence, FE formulation becomes convenient:

- Break the structure into separate simple elements and define stiffness matrix for all elements, K^e .
- Assemble elements into a global model
- Solve for motion at element ends or "nodes", d^e .
- $u(x)$ approximation gives motion within each element
- Motion in each element $\rightarrow \sigma, \varepsilon$ as functions of x along each element ($\sigma(x), \varepsilon(x)$).

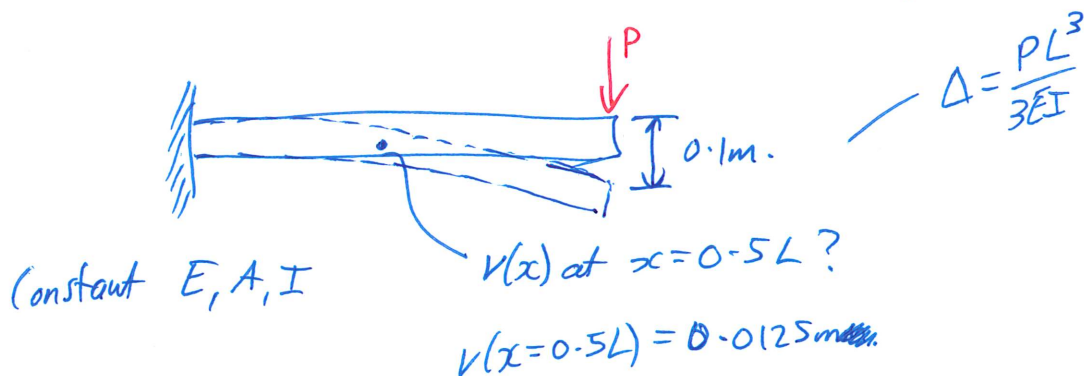
The goal is to use "simple" elements to create solutions to complex problems where analytical methods fail (or become too complex to solve easily).



Constant N
 Constant E
 Constant A .



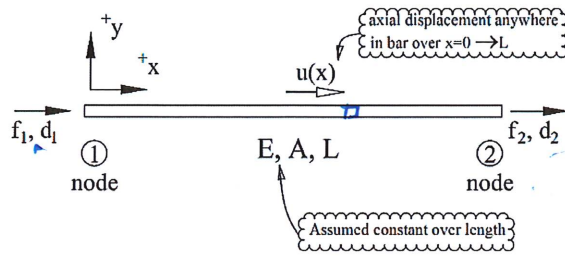
Internal element actions are inferred from values at the nodal points.



$$[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{f\}$$

Bar elements

f_1 and f_2 must be equal and opposite for the bar to be in static equilibrium.



f_1, f_2 - nodal forces at nodes 1 and 2, respectively
 d_1, d_2 - nodal displacements at nodes 1 and 2, respectively

(1) Note: 2 DOF (1 at each end of element)

(2) Goal: to be able to define/find $u(x)|_{x=0 \rightarrow L}$ based on d_1 and d_2 .

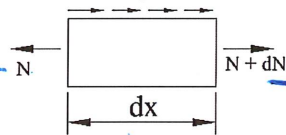
(3) Consider an infinitesimally small section of bar:

A way of defining continuous internal element mechanics based upon only the values at the end points.

p = distributed axial force within the bar.

$p \cdot dx$ = axial force internal to the bar

N = internal normal axial force within this section of bar.



the change in internal normal force over length dx .

$$\sum F_x = 0 = -N + (N + dN) + p \, dx$$

$$-dN = p \, dx$$

$$-\frac{dN}{dx} = p$$

(4) We know from ENME202 $\rightarrow N = \sigma A = (E\epsilon)A = EA \frac{du}{dx}$

$\frac{du}{dx}$ = the rate of change of displacement with respect to position within the bar (strain)

(5) Hence, substituting Equation (4) to Equation (3)

$$-\frac{dN}{dx} = -\frac{d}{dx} \left(EA \frac{du}{dx} \right) = p(x)$$

the rate of change of internal normal force.

$p(x)$ distributed axial load intensity.

(6) Governing Boundary Value Problem (BVP):

Given: A, E, f_1 and f_2 over element $x = 0 \rightarrow L$

Find: $u(x)$ such that:

$$-\frac{d}{dx} \left(EA \frac{du}{dx} \right) = 0$$

$$u(x = 0) = d_1$$

$$u(x = L) = d_2$$

Equilibrium of a "small" element, based upon the homogeneous form, with $p(x) = 0$

Some as yet undefined equation linking internal behaviour to nodal values.

Called the "strong form" of problem, where the exact solution is hard. This form requires the conditions above to be true for all the element, from $x = 0$ to $x = L$.

(7) Homogenous strong form in (6) assumes the distributed axial load acting between nodes is equal to zero, $p(x) = 0$. This means that the bar carries a constant axial force along its length.

(8) Therefore, to solve (6) assuming, E, A are constant, we define a solution:

$$u(x) = a_0 + a_1 x$$

Where a_0 and a_1 are determined from boundary conditions (BC's)

$$\begin{cases} u(x = 0) = d_1 \\ u(x = L) = d_2 \end{cases}$$

$$a_0 + a_1 \cdot 0 = d_1 \rightarrow a_0 = d_1$$

$$d_2 = d_1 + a_1 L \rightarrow a_1 = \frac{d_2 - d_1}{L}$$

Hence, the function describing displacement of element anywhere along $0 \rightarrow L$ can be written:

$$u(x) = d_1 + \frac{d_2 - d_1}{L} x$$

$$u(x) = \left(1 - \frac{x}{L}\right) d_1 + \frac{x}{L} d_2$$

$$u(x) = \psi_1(x) d_1 + \psi_2(x) d_2$$

Constant A
Constant E
Constant N
distributed axial load $p(x) = 0$.

straight line equation (compare to $y = mx + c$)

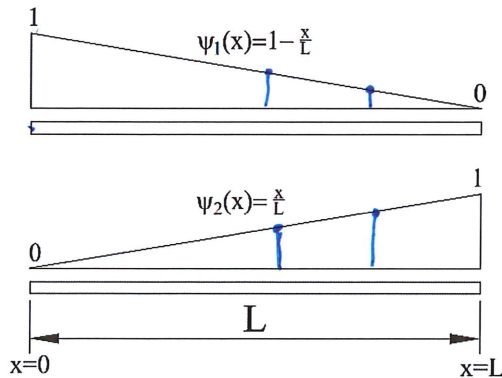
Our choice of the form of this equation is based upon our prior knowledge of how the element will deflect. It also defines limitations in the scope of problems that can be solved using this element type.

$$\psi_1(x) = 1 - \frac{x}{L}$$

$$\psi_2(x) = \frac{x}{L}$$

(9) Where ψ_1, ψ_2 are shape functions describing change in $u(x)$ between nodes. In other words, these shape functions interpolate between nodal displacements d_1 and d_2 .

Shape functions ψ_1 and ψ_2 can be defined *plotted:*



These shape functions can be thought of like "participation factors" or "weighting factors".

$$u(x) = \left(1 - \frac{x}{L}\right) d_1 + \frac{x}{L} d_2$$

$$u(x) = \psi_1(x) d_1 + \psi_2(x) d_2$$

Example:

Distance, x	$x = 0.5L$	$x = 0.75L$
Displacement within the bar, $u(x)$	$u(x = 0.5L) = 0.5d_1 + 0.5d_2$	$u(x = 0.75L) = 0.25d_1 + 0.75d_2$
Strain, $\varepsilon(x)$	$\varepsilon(x) = \frac{du}{dx} = \frac{(-d_1 + d_2)}{L}$ Strain $\varepsilon(x)$ is independent of x, hence $\varepsilon(x) = \text{const} = \frac{(-d_1 + d_2)}{L}$	
Stress, $\sigma(x)$	$\sigma(x) = E\varepsilon = \frac{E(d_2 - d_1)}{L} = E \frac{\Delta L}{L}$	

Equation (9) defines displacement for $x = 0 \rightarrow L$. To get a stiffness matrix relating element forces f to the element displacements d (where $f = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$ and $d = \begin{pmatrix} d_1 \\ d_2 \end{pmatrix}$) we need forces along $x = 0 \rightarrow L$.

Recall, Equation (4) $N = EA \frac{du}{dx}$ using (9):

$$N = EA \frac{d}{dx} \left(\left(1 - \frac{x}{L}\right) d_1 + \left(\frac{x}{L}\right) d_2 \right)$$

$$N = \frac{EA}{L} (d_2 - d_1) = \frac{EA}{L} \Delta L = EA \left(\frac{\Delta L}{L} \right) = EA \epsilon = (\epsilon E) A = \sigma A.$$

Hence axial force, N , is constant along $x = 0 \rightarrow L$ as expected with distributed axial load intensity, $p(x) = 0$ for the homogenous form.

Now relate axial force, N , with nodal forces f_1 and f_2

node 1

$$\sum F_x = 0 = f_1 + N(0)$$

$$f_1 = -N_1(0) = -\frac{EA}{L} (d_2 - d_1)$$

$$\underline{f_1 = \frac{EA}{L} (d_1 - d_2)}$$

node 2

$$\sum F_x = 0 = f_2 - N(L)$$

$$f_2 = N(L)$$

$$\underline{f_2 = \frac{EA}{L} (d_2 - d_1)}$$

(13) Combine into matrix form:

Element formulation for an axial bar element

$$\begin{pmatrix} -N(0) \\ N(L) \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix}$$

$$f^e = K^e d^e$$

2 unknowns
2 equations
We can't solve this.

Normally, this solvable, but the 2 equations are not independent.

stiffness matrix, K^e
for the bar element
that relate applied
loads to corresponding deflections.

Matrix
determinant

(14) **Note:** K^e is singular $\det(K^e) = 0$

This means that $f^e = K^e d^e$ has no unique solution b/c it is not constrained

Hence: $d_1 = d_2 = 10 \rightarrow d^e = \begin{pmatrix} d_1 \\ d_2 \end{pmatrix}$

$$f^e = K^e d^e = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{pmatrix} 10 \\ 10 \end{pmatrix} = \frac{EA}{L} \begin{pmatrix} 10 - 10 \\ -10 + 10 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

You get motion without force when a structure is not constrained. This holds for any nodal displacement values where $d_1 = d_2$.

(15) Questions

a) What happens if E and A varies over $x = 0 \rightarrow L$

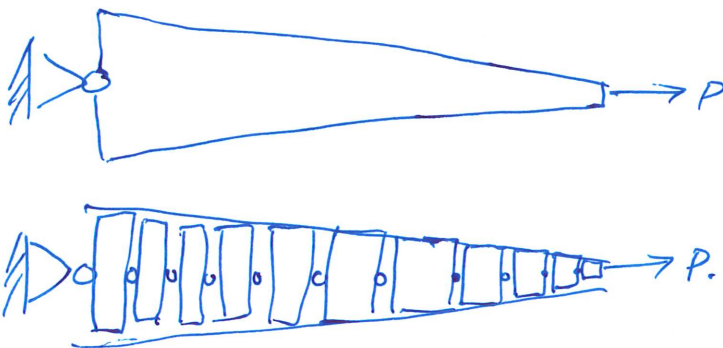
- Need a new element
- Or to use many elements

b) What happens if $p(x) \neq 0$?

- You get

$$-\frac{d}{dx} \left(EA \frac{du}{dx} \right) = p(x) \neq 0$$

- Get a new element form
- Or you see later in the notes



Shape Functions:

Shape functions have two key purposes:

1. To enable a system of equations to be developed so that solving for displacements calculated at only a few selected discrete points within the system (the nodal points) can represent the full, continuous structural system.
 2. Once these displacements are solved, we can then use the shape functions and the solved nodal deflections (and element geometric and material properties) to calculate internal deflections, forces, stresses and strains within the element (not just at the nodal points).
- ➔ Shape functions are used to both “build-up” and “break down” or interpret the structure.

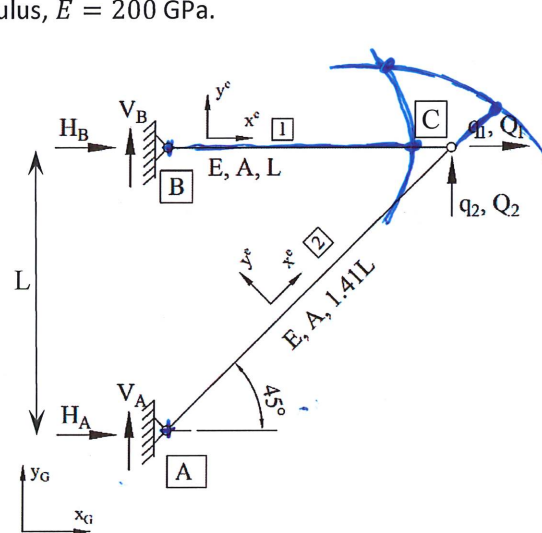
A brief review of energy methods and virtual displacements

You will (hopefully) recall the use of energy methods to solve displacements in structures. We will briefly re-cap Energy Methods, the Work-Energy Method for Single Loads, and the Method of Virtual Work.

First up, we'll consider a simple system of axial bars. We will solve this first through a manual method as we might have done last year in ENME202. We will then solve it through the Method of Virtual Work, but do this manually. Observing this process in detail for a given example will set the scene for the more general system that we develop to solve a range of problems.

Simple Axial Bar Problem

External forces $Q_1 = 0$, $Q_2 = 100\text{kN}$ act on the structure below. Find the structural deflections at pin-joint C. Assume $L_1 = 10\text{m}$ and $L_2 = 14.1\text{m}$, and both members are made of 100mm diameter solid steel bars with elastic modulus, $E = 200\text{ GPa}$.



At a structural level, we use lowercase q to define allowable degrees of freedom and uppercase Q to define the corresponding applied external loads.

Solution – Using simple hand calculation for statically determinate systems

We will first consider force equilibrium at the top right pin joint:

Vertical Force Equilibrium: $\sum F_y = 0: 100\text{kN} - F^2 \cos(45^\circ) = 0$

$$F^2 = \frac{100\text{kN}}{\cos(45^\circ)} = 141.42\text{kN}$$

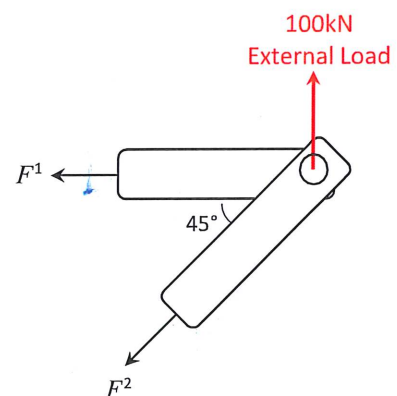
Horizontal Force Equilibrium: $\sum F_x = 0: -F^1 - F^2 \cos(45^\circ) = 0$

$$F^1 = -F^2 \cos(45^\circ) = -100\text{kN}$$

Calculate elongation of each element based upon the internal loads.

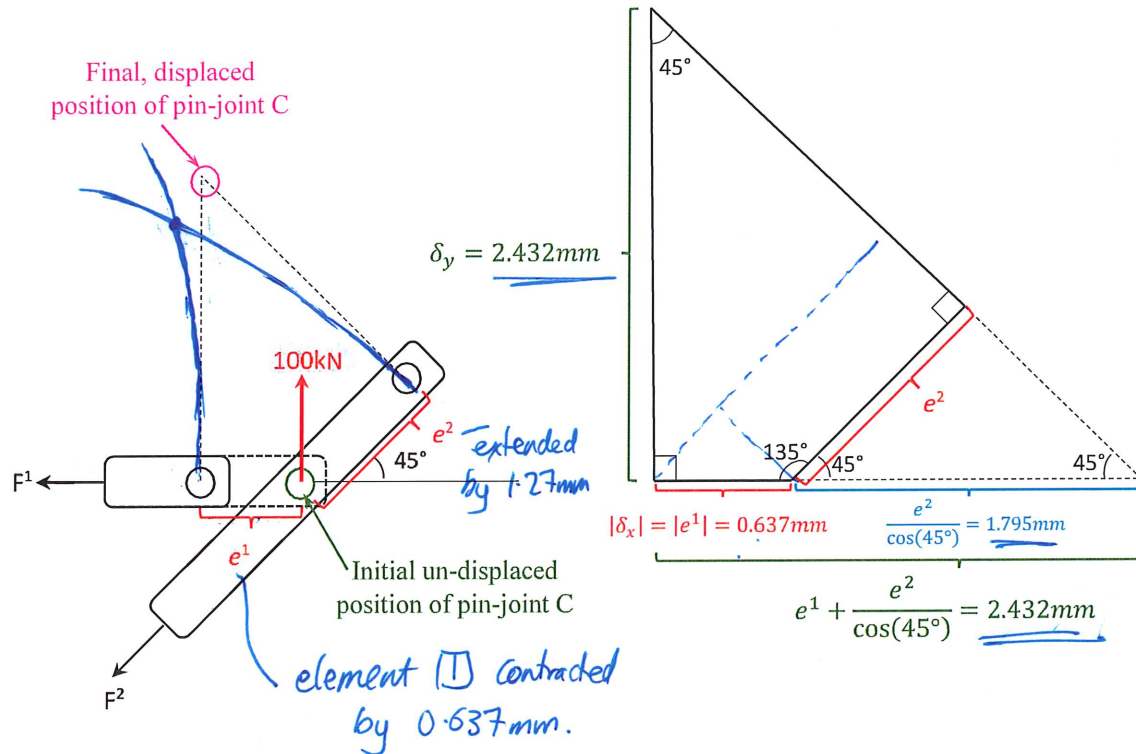
Member 1: $e^1 = \frac{F^1 L^1}{A^1 E^1} = \frac{(-100,000)(10)}{\left(\frac{\pi}{4} 0.1^2\right)(200 \times 10^9)} = -6.36619 \times 10^{-3}\text{m} = -0.637\text{mm}$

Member 2: $e^2 = \frac{F^2 L^2}{A^2 E^2} = \frac{(141,420)(14.1)}{\left(\frac{\pi}{4} 0.1^2\right)(200 \times 10^9)} = 1.2694 \times 10^{-3}\text{m} = 1.27\text{mm}$



Now, we need to manually interpret these individual element deflections to determine the displacements between the initial and final locations of pin-joint C.

After some trigonometry and messing about with an awkward quadrilateral shape, we can determine that pin-joint C displaces horizontally to the left by 0.637mm and vertically upwards by 2.432mm.



A simple, statically-determinate structure like this can be solved by hand, but it gets complicated quickly and we would like a more general approach. The first such approach that we will consider is the Work-Energy Method for Single Loads.

Point C moves by 0.637mm to the left horizontally

and upwards by 2.432mm vertically.

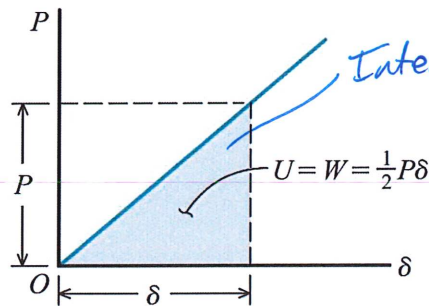
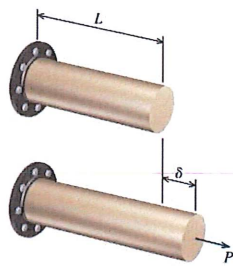
$W = F\delta$

This equation is only valid for constant force.

Work-Energy Method for Single Loads

When a load is applied and a body deforms, work is done on the body. The **external forces** are said to do **external work**. The **internal reactions** to these loads do **internal work**, often referred to as **strain energy**. Think about strain energy like the potential energy stored in a stretched/strained spring.

Work of a force: Work is defined as the product of a force times the distance that it moves, in the direction of that force.



$W = \frac{1}{2} P\delta$

external work done

The principle of conservation of energy states that energy is neither created nor destroyed, rather transformed from one state to another. Therefore, the work done by the external force is stored within the material as strain energy. Provided no energy is lost as heat, then we can say that the strain energy, U , is equal in magnitude to the external work, W :

$$U = W = \int_0^{\delta_1} P d\delta$$

For axial deformation, we have the equation for internal strain energy as:

$$U = \frac{1}{2} P\delta$$

For prismatic bars, with constant cross-section, $\delta = PL/AE$, giving an expression for strain energy in terms of applied force, P :

$$U = \frac{P^2 L}{2AE}$$

The Work-Energy Method for Single Loads can be used to determine the deformations of a structural member for **very selected conditions**:

- The member or structure must be loaded by a single external concentrated force
- Corresponding displacements can only be determined at the location of the single load and in the direction that the load acts

Why are we restricted to a single external load? The equation above ($W = U$) is the only equation we have available to us. The strain energy, U , will be a single number. The work, W , performed by the external load is also only a single number.

Consequently, if more than one external force is applied, then the term for W will have more than one unknown deflection or rotation angle.

→ One equation, two (or more) unknowns = not solvable, without extending the methodology.

Simple Axial Bar Problem – Solved using the Work-Energy Method for Single Loads

External forces $Q_1 = 0$, $Q_2 = 100\text{kN}$ act on the structure below. Find the structural deflections at pin-joint C. Assume $L_1 = 10\text{m}$ and $L_2 = 14.1\text{m}$, and both members are made of 100mm diameter solid steel bars with elastic modulus, $E = 200\text{ GPa}$.

The first steps are the same as before. We consider force equilibrium at the top right pin joint:

Vertical Force Equilibrium: $\sum F_y = 0: 100\text{kN} - F^2 \cos(45^\circ) = 0$

$$F^2 = \frac{100\text{kN}}{\cos(45^\circ)} = \underline{\underline{141.42\text{kN}}}$$

Horizontal Force Equilibrium: $\sum F_x = 0: -F^1 - F^2 \cos(45^\circ) = 0$

$$F^1 = -F^2 \cos(45^\circ) = \underline{\underline{-100\text{kN}}}$$

Based upon these solved internal forces, we can calculate the internal strain energy within the structure.

Member 1: $U^1 = \frac{(F^1)^2 L^1}{2A^1 E^1} = \frac{(-100,000)^2 (10)}{2 \left(\frac{\pi}{4} 0.1^2\right) (200 \times 10^9)} = \underline{\underline{31.83\text{ J}}}$

Member 2: $U^2 = \frac{(F^2)^2 L^2}{2A^2 E^2} = \frac{(141,420)^2 (14.1)}{2 \left(\frac{\pi}{4} 0.1^2\right) (200 \times 10^9)} = \underline{\underline{89.76\text{ J}}}$

The total internal strain energy is: $U^{\text{Total}} = U^1 + U^2 = 31.83\text{ J} + 89.76\text{ J} = 121.59\text{ J}$

Equating external work to internal strain energy: $U^{\text{Total}} = W^{\text{External}} = \frac{1}{2} P \delta_y = 121.59\text{ J}$

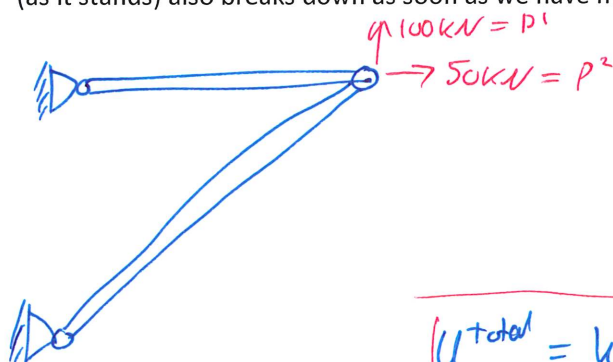
And we can then solve for the vertical deflection at the point that the load is applied:

$$\delta_y = \frac{2U^{\text{Total}}}{P} = \frac{2(121.59)}{100,000} = 2.432 \times 10^{-3}\text{m} = 2.432\text{mm}$$

rearrange and make δ_y the subject.

This solved vertical deflection is the same as result that we obtained before, but without all the manual intervention, without the awkward quadrilateral, and without the pesky triangles and trigonometry.

However, this method (as it stands) cannot give us the horizontal deflection at pin-joint C. This method (as it stands) also breaks down as soon as we have more than one load acting on the structure.

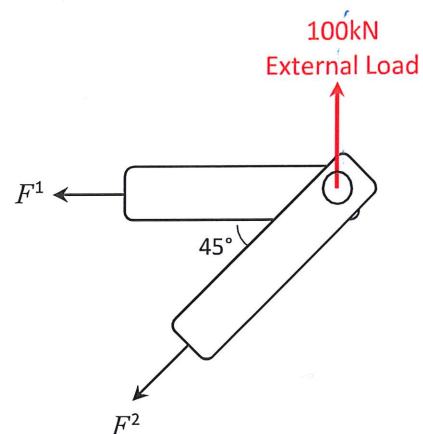
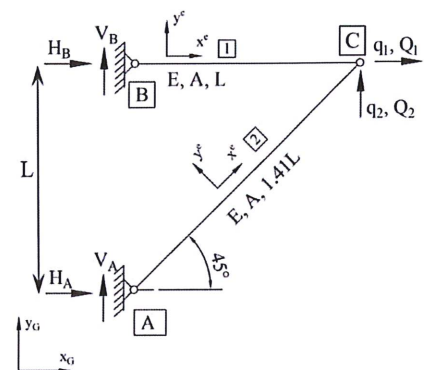


$$U^{\text{Total}} = \frac{(F^1)^2 L^1}{2A^1 E^1} + \frac{(F^2)^2 L^2}{2A^2 E^2}$$

$$W^{\text{ext}} = \frac{1}{2} P^1 \delta_y + \frac{1}{2} P^2 \delta_x$$

$$U^{\text{total}} = W^{\text{ext}}$$

If we apply multiple external loads, we have multiple unknowns, a single equation and the method fails.



Method of Virtual Work

The Method of Virtual Work is an important extension of the Work-Energy Method for Single Loads. The Principle of Virtual Work allows us to solve for deflections for multiply-loaded structures, and at locations other than those where external loads are applied. The extension to the method is actually relatively simple to implement, but can be difficult to understand conceptually.

The Principle of Virtual Work states:

If a deformable body is in equilibrium under a virtual force system and remains in equilibrium while it is subjected to a set of small, compatible deformations, then the external virtual work done by the virtual external forces acting through the real external displacements (or rotations) is equal to the virtual internal work done by the virtual internal forces acting through the real internal displacements (or rotations).

But what does that actually mean? That statement made my brain hurt...

Procedurally, if we have a structure subjected to a single-load and we want the deflection at a different location to that of the load (or at that point but not in the direction of that load), or for multiply-loaded structures, then we can apply a virtual external load.

We use the principle of superposition to independently consider the real loads and the virtual loads and how each load is developed within the structure.

We typically define the virtual external load to have a value of 1.0 (N or kN) to make life easier. We have to choose something and choosing this value makes the calculations a little bit simpler.

When used to analyse a truss structure, this process has several steps:

- First, take the real external loads applied only (ignore the virtual load for now) and work out the internal forces developed within each member.
- Then, ignore the real external loads and only consider the chosen virtual external load applied to the structure. This virtual load is applied at the point that you wish to know the deflection and acts in the direction in which you wish to know the deflection. Work out the internal forces that would be induced by the application of this virtual external load (we call these the virtual internal forces).
- Apply the Principle of Virtual Work and equate the Virtual Internal Work (the work done by the internal virtual forces moving through the real displacements) to the Virtual External Work (the work done by the virtual external load moving through the real external displacement).

The principle can be expressed in an equation as:

$$\begin{matrix} \text{Virtual} \\ \text{External} \\ \text{Loads} \end{matrix} \times \begin{matrix} \text{Real External} \\ \text{Displacements} \end{matrix} = \sum \left(\begin{matrix} \text{Virtual} \\ \text{Internal} \\ \text{Forces} \end{matrix} \times \begin{matrix} \text{Real Internal} \\ \text{Displacements} \end{matrix} \right)$$

For a compound truss with homogenous (constant elastic modulus) and prismatic (constant cross-section) members, this principle can be expressed as:

$$1 \cdot \Delta = \sum_i f^i \left(\frac{F^i L^i}{A^i E^i} \right)$$

where:

1 = virtual external unit load acting at the location we desire to obtain deflection, Δ , and in the direction in which we want to solve for deflection.

Δ = real joint displacement caused by the real loads that act on the truss.

f^i = virtual internal force created within truss member i when the truss is loaded with only the single external virtual load (ignoring any real external loads that act on the truss)

F^i = real internal force created within truss member i when the truss is loaded with all the real loads (ignoring the virtual load)

L^i = the length of truss member i

A^i = the cross-sectional area of truss member i

E^i = the elastic modulus of truss member i

Simple Axial Bar Problem – Solved using the Method of Virtual Work

External forces $Q_1 = 0$, $Q_2 = 100\text{kN}$ act on the structure below. Find the structural deflections at pin-joint C. Assume $L_1 = 10\text{m}$ and $L_2 = 14.1\text{m}$, and both members are made of 100mm diameter solid steel bars with elastic modulus, $E = 200\text{ GPa}$.

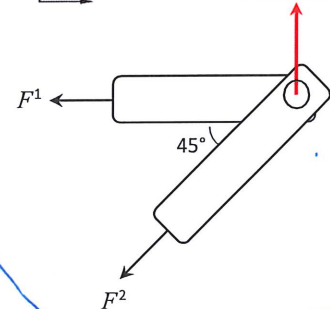
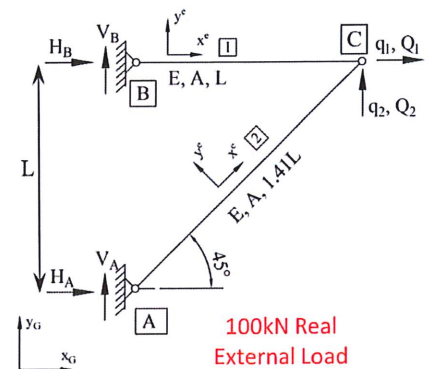
The first steps are the same as before. We consider force equilibrium at the top right pin joint:

Vertical Force Equilibrium: $\sum F_y = 0: 100\text{kN} - F^2 \cos(45^\circ) = 0$

$$F^2 = \frac{100\text{kN}}{\cos(45^\circ)} = 141.42\text{kN}$$

Horizontal Force Equilibrium: $\sum F_x = 0: -F^1 - F^2 \cos(45^\circ) = 0$

$$F^1 = -F^2 \cos(45^\circ) = -100\text{kN}$$



We wish to determine the horizontal deflection at pin-joint C, so we will now ignore the real externally applied load and then apply a unit virtual load in the horizontal direction at C. The corresponding internal loads within each member are the virtual internal forces.

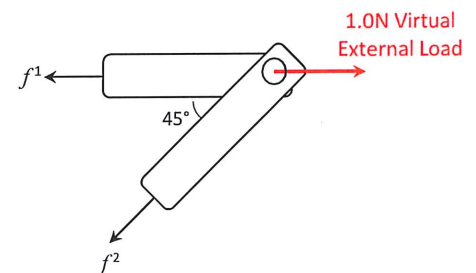
Vertical Force Equilibrium: $\sum f_y = 0: -f^2 \cos(45^\circ) = 0$

$$f^2 = \frac{-0}{\cos(45^\circ)} = 0\text{N}$$

Horizontal Force Equilibrium:

$$\sum F_x = 0: -f^1 - f^2 \cos(45^\circ) + 1.0\text{N} = 0\text{N}$$

$$f^1 = 1.0 - f^2 \cos(45^\circ) = 1.0\text{N}$$



Member	Length, L (m)	Area, A (m ²)	Elastic Modulus, E (Pa)	Real Internal Force, F (N)	Virtual Internal Force, f (N)	$f^i \left(\frac{F^i L^i}{A^i E^i} \right)$ (kN ² m)
1	10	7.854×10^{-3}	200×10^9	-100,000	1.0	-6.366×10^{-4}
2	14.1	7.854×10^{-3}	200×10^9	141,421	0	0
$\sum_i f^i \left(\frac{F^i L^i}{A^i E^i} \right)$						-6.366×10^{-4}

$$1 \cdot \Delta = \sum_i f^i \left(\frac{F^i L^i}{A^i E^i} \right) \rightarrow \Delta = \sum_i f^i \left(\frac{F^i L^i}{A^i E^i} \right) = -6.366 \times 10^{-4}\text{m} = -0.6366\text{mm}$$

What if instead, we had wanted the vertical displacement (like we did earlier), what would that look like?

The first part where we calculate the real internal force does not change (the real loads have not changed in this instance), so we will not repeat that part.

We now wish to determine the **vertical** deflection at pin-joint C, so we again ignore the real externally applied load and apply a unit virtual load, this time in the **vertical** direction at C. The corresponding internal loads within each member are the virtual internal forces.

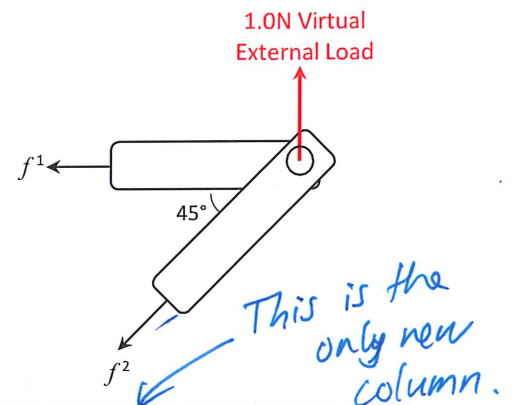
Vertical Force Equilibrium: $\sum f_y = 0: -f^2 \cos(45^\circ) + 1.0 = 0$

$$f^2 = \frac{1.0}{\cos(45^\circ)} = \underline{1.4142N}$$

Horizontal Force Equilibrium:

$$\sum F_x = 0: -f^1 - f^2 \cos(45^\circ) = 0N$$

$$f^1 = -f^2 \cos(45^\circ) = \underline{-1.0N}$$

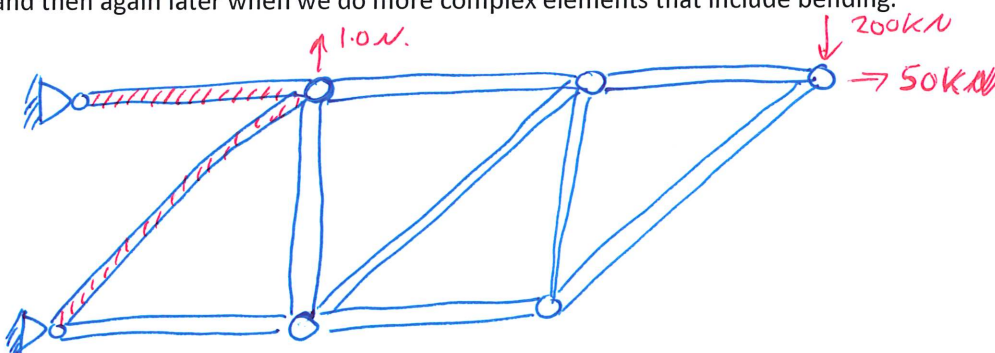


Member	Length, L (m)	Area, A (m ²)	Elastic Modulus, E (Pa)	Real Internal Force, F (N)	Virtual Internal Force, f (N)	$f^i \left(\frac{F^i L^i}{A^i E^i} \right)$ (kN ² m)
1	10	7.854×10^{-3}	200×10^9	-100,000	-1.0	6.366×10^{-4}
2	14.1	7.854×10^{-3}	200×10^9	141,421	1.4142	1.795×10^{-3}
$\sum_i f^i \left(\frac{F^i L^i}{A^i E^i} \right)$						<u>2.432×10^{-3}</u>

$$1 \cdot \Delta = \sum_i f^i \left(\frac{F^i L^i}{A^i E^i} \right) \rightarrow \Delta = \sum_i f^i \left(\frac{F^i L^i}{A^i E^i} \right) = 2.432 \times 10^{-3}m = \underline{2.432mm}$$

This simple example will hopefully give you some confidence with the theory behind the principle of virtual displacements and how they are applied. This structure is simple and statically-determinate, so we can easily solve for real and virtual internal forces within each structural member. However, if we want to apply this to more complex statically-indeterminate structures, then we will need to develop a better/more extensive computational framework to solve such problems.

The principle of virtual displacements is used in the derivation of our stiffness matrices for bar elements, and then again later when we do more complex elements that include bending.



Derive the element stiffness matrix in a different (and simpler) way

(16) Bar element formulation can be obtained in a simpler way using the Principle of Virtual Displacements (PVD). Given Equation (6):

$$-\frac{d}{dx}\left(EA \frac{du}{dx}\right) = 0$$

Using force boundary conditions:

$$\begin{aligned} -EA \frac{du}{dx} \Big|_{x=0} &= f_1 \\ EA \frac{du}{dx} \Big|_{x=L} &= f_2 \end{aligned}$$

(17) Multiply both sides of Equation (16) by virtual (small) displacement, δu :

$$\delta u \left(-\frac{d}{dx} \left(EA \frac{du}{dx} \right) \right) = \delta u(0) = 0 \quad \forall \quad x = 0 \rightarrow L$$

(18) Integrate over element domain $\Omega = (0 \rightarrow L)$ which is a less strict requirement than Equation (17) above:

$$\int_0^L \delta u \left(-\frac{d}{dx} \left(EA \frac{du}{dx} \right) \right) dx = \int_0^L \delta u \left(EA \frac{d^2 u}{dx^2} \right) dx = 0$$

And integrate by parts so that u and δu have the same order of differentiation (currently δu – zero order and u is second order):

Integrate by parts

$$\int_0^L \delta u \left(EA \frac{d^2 u}{dx^2} \right) dx = 0$$

$$\left[\delta u \left(EA \frac{du}{dx} \right) \right]_0^L - \int_0^L \frac{d}{dx} \delta u \left(EA \frac{du}{dx} \right) dx = 0$$

Recall the formula for integration by parts

$$\int g(x) h'(x) dx = g(x) h(x) - \int h(x) g'(x) dx$$

In this instance, we will define:

$$g(x) = \delta u \quad g'(x) = \frac{d}{dx} \delta u$$

$$h(x) = EA \frac{du}{dx} \quad h'(x) = EA \frac{d^2 u}{dx^2}$$

$$\delta u(L) EA \frac{du}{dx} \Big|_{x=L} - \delta u(0) EA \frac{du}{dx} \Big|_{x=0} - \int_0^L \frac{d}{dx} \delta u \left(EA \frac{du}{dx} \right) dx = 0$$

And after applying boundary conditions above:

$$\delta u(L) f_2 - \delta u(0) (-f_1) - \int_0^L (\delta u') \left(EA \frac{du}{dx} \right) dx = 0$$

$$\boxed{\int_0^L (\delta u') \left(EA \frac{du}{dx} \right) dx = \delta u(0) (f_1) + \delta u(L) f_2}$$

It's called the "weak form" because it must be true over integrated terms only.

Compared with "strong form" which must be true for all x .

Uppercase Ψ - ~~a~~ a vector of axial shape functions
 lowercase ψ - individual shape functions

(19) Now we use given or known shape functions, $\Psi(x) = [\psi_1(x) \ \psi_2(x)]$, to substitute for:

$$u(x) = \Psi(x) d = \psi_1 d_1 + \psi_2 d_2 \quad - \quad [\psi_1(x) \ \psi_2(x)] \begin{Bmatrix} d_1 \\ d_2 \end{Bmatrix}$$

$$\delta u(x) = \Psi(x) \delta d = \psi_1 \delta d_1 + \psi_2 \delta d_2$$

Or,

$$u(x)^T = d^T \Psi(x)^T = \psi_1 d_1 + \psi_2 d_2$$

$$\delta u(x)^T = \delta d^T \Psi(x)^T = \psi_1 \delta d_1 + \psi_2 \delta d_2$$

where $\psi_1 = (1 - \frac{x}{L})$; $\psi_2 = \frac{x}{L}$ are pre-defined interpolation/shape functions.

(20) Equation (18) becomes:

$$\int (\delta d^T \Psi'(x)^T) EA (\Psi'(x) d) dx = \delta d^T f = \delta d^T \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$$

where $\Psi'(x) = \frac{d}{dx}(\Psi(x)) = (\frac{d}{dx} \psi_1 \ \frac{d}{dx} \psi_2)$

$$\delta d^T \int (\Psi'(x)^T EA \Psi'(x) dx) d = \delta d^T f$$

$$\int (\Psi'(x)^T EA \Psi'(x) dx) d = f$$

look at the form of this equation.
 (.....) $d = f$.

All of that stuff in the brackets is stiffness.

$$K^e d^e = f^e$$

$$K^e = \int_0^L (\Psi'(x)^T EA \Psi'(x) dx)$$

$$\text{where } \Psi' = \begin{pmatrix} -\frac{1}{L} & \frac{1}{L} \end{pmatrix}$$

The equations provide a definition of the relationship between applied loads and deflections.
 A definition of stiffness based upon our chosen shape functions.

(21) Deriving the final form of bar element stiffness matrix

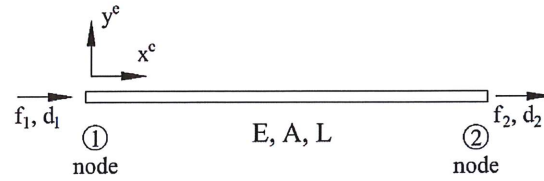
$$K^e = \int_0^L \begin{pmatrix} -\frac{1}{L} \\ \frac{1}{L} \end{pmatrix} EA \begin{pmatrix} -\frac{1}{L} & \frac{1}{L} \end{pmatrix} dx = \frac{EA}{L^2} \int_0^L \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} dx$$

$$= \frac{EA}{L^2} \left(\begin{bmatrix} L & -L \\ -L & L \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right)$$

$$K^e = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Just as before...but this derivation makes K^e easy to calculate given shape functions $\psi(x)$ to relate or interpolate motion along the element from nodal (end) displacement values.

Summary



Axial bar element stiffness matrix derived:

$$f_{\{2 \times 1\}}^e = K_{\{2 \times 2\}}^e d_{\{2 \times 1\}}^e$$

$$f^e = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} \quad d^e = \begin{pmatrix} d_1 \\ d_2 \end{pmatrix}$$

$$K^e = \int_0^L (\Psi'(x))^T EA \Psi'(x) dx = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$K_{\{2 \times 2\}}^e = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\begin{Bmatrix} f_1^e \\ f_2^e \end{Bmatrix} = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} d_1^e \\ d_2^e \end{Bmatrix}$$

Element stiffness matrix, K^e , is a function of E , A , L , and the given shape functions.

Axial forces along bar assumed to be constant, ie: $p(x) = 0$.

Captures only axial deformation and force displacement behaviour. It does not carry moments and has no rotations.

Stiffness matrix is derived based on shape functions:

$$\Psi = [\psi_1(x) \quad \psi_2(x)]$$

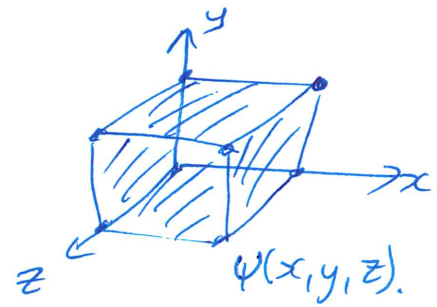
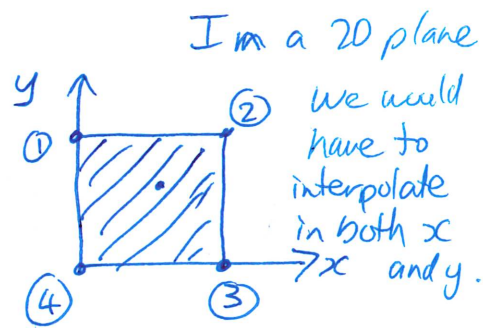
$$\psi_1(x) = 1 - \frac{x}{L}; \quad \psi_2(x) = \frac{x}{L}$$

Bar element axial deformation at any point x can be estimated:

$$u(x) = \psi_1(x)d_1 + \psi_2(x)d_2 = \Psi d^e$$

Bar element coordinate axes are defined by (x^e, y^e) where x^e is aligned along element and y^e perpendicular to the longitudinal axis of the element.

Final notes:



- Bars (and beams) are 1D elements, so shape functions are along only that 1 dimension- i.e. $\psi = \text{function of } x, \psi(x)$, and not $\psi(x, y)$ if we had a plate instead.
- This process is very general. Given a governing ODE and boundary conditions as well as known exact solution or estimated shape functions, $u = \Psi d$, then this process holds in general for a very wide range of finite element types.
- Hence, knowing this approach will allow you to understand (almost) any finite element you might encounter now in a software package (or in future).
- Recall though that our governing ODE in this axial bar case only was for “small” deflections and thus, our K^e matrix and $f^e = K^e d^e$ is also only valid for small deflections.

→ It is assumed that the initial geometry of the structure is a fair approximation of the final geometry ie: that the deflected structure doesn't have significantly different geometry than the undeflected structure.

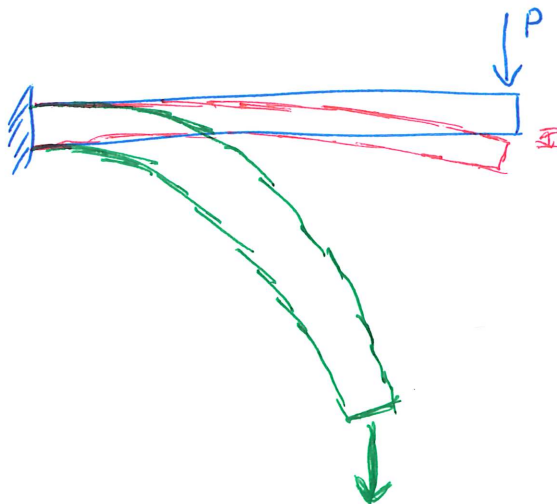
→ Consequently, it is assumed that the stiffness matrix built up from the initial geometry is an accurate mathematical representation of the overall structural system, in both the undeflected and deflected position.

What if I want to model a structure that undergoes large deformations?

→ Build up a structural model based upon the initial geometry.

→ Use the approximated deflected shape to build up a new stiffness matrix that is based on the approximation of the deflected shape.

→ Iterate on this process until you get a converged solution.



Questions

- 1) How do I make a model of something real?
- 2) What if bar element axes (x^e, y^e) do not match the global model reference coordinates?
- 3) What if elements do not all line up in coordinate system?

Answer

Model an assembly of a series of element stiffness matrices into a full model.

Important Terminology for extending to multiple-element structures:

We will define global (structure-level) degrees of freedom which indicate the permissible ways in which a structure can move and deflect. Any position where there is the potential for a non-zero displacement to occur at a nodal point, we assign a global degree of freedom at that location, and in the direction in which there is the potential for displacement.

q – a permissible degree-of-freedom where there is the potential for non-zero displacements to occur. The degrees of freedom **q** that we assign always align with our global coordinates, X^G and Y^G .

Q – an externally applied load (which may be zero) that exists at this nodal point. We always assign a degree of freedom **q** and external forcing term **Q** as a matching pair. Even if there is no externally applied load at this nodal point, we still assign the variable and just make the value zero.

If a nodal point is constrained by an external support such as a pin or roller, we do not assign a **q** value in a direction if the constraint means that the node cannot move in this direction.

